

AI-Integrated Coding Environments

We categorize AI-assisted coding tools into **CLI tools**, **VS Code (and IDE) extensions**, and **AI-native editors/environments**. The tables below compare key features (models, open-source status, workflow support, etc.) for representative tools in each category. Each tool's entry cites its documentation or analysis for verification.

CLI Tools

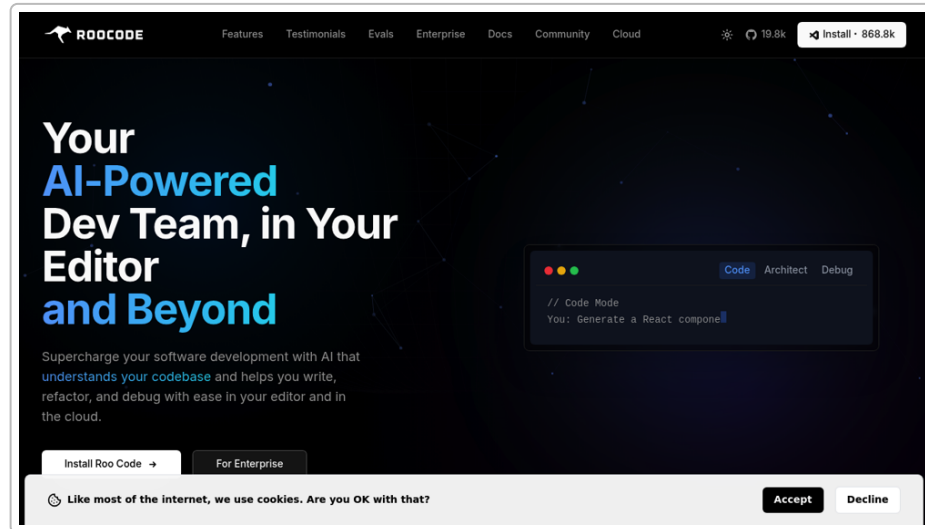
Tool	Source	Models	BYO Model	Key Features	Multi-file Edits	Agentic Tasks	IDE/ Terminal
Gemini CLI	OpenSource ¹	Google Gemini 2.5 Pro	Yes (MCP)	ReAct loop agent; Google Search integration; scripting ¹ ²	✓	Yes	Terminal (any OS)
Claude Code	Closed (Anthropic) ³	Anthropic Claude 3.7 Sonnet; Opus/Haiku (OpenAI)	Partially (via Bedrock/Vertex)	Codebase understanding; GitHub/GitLab integration; permissioned changes ³ ⁴	✓	Yes	Terminal + Git CLI
Copilot CLI	Closed (GitHub) ⁵	GitHub Copilot engine (multimodel)	No (GitHub managed)	Agent knows repo/PR context; runs commands; edits code with approval ⁵ ⁶	✓	Yes	Terminal (via <code>@github/copilot</code>)
Codex CLI	Open (OpenAI) ⁷	GPT-5-Codex (OpenAI)	Yes (ChatGPT API key)	TUI agent; reads/edits/ runs code locally; image & web search support ⁷ ⁸	✓	Yes	Terminal (Rust-based)

Tool	Source	Models	BYO Model	Key Features	Multi-file Edits	Agentic Tasks	IDE/ Terminal
Aider	Open (GitHub) ⁹	OpenAI GPT-4, Claude Sonnet, DeepSeek, etc.	Yes (any LLM, incl. local)	CLI pair programmer; supports 100+ languages; Git integration; live testing ⁹	✓	No	Terminal
Augment Code	Closed (Augment) ¹⁰	Frontier LLMs (cloud)	No (cloud service)	Autonomous agents (cloud/ IDE); codebase index; image context; TUI/ CLI; runs commands ¹⁰ ¹¹	✓	Yes	Terminal + IDE plugins
Kilo Code	Open ¹²	Any (500+ via OpenRouter)	Yes (BYOK/ OpenRouter/ local)	Multi-agent modes (Orchestrator, Debug, etc.); remembers context; auto-debug/ recovery ¹³ ¹⁴	✓	Yes	CLI & IDE (VSCode/ JetBrains) ¹³
Kiro	Closed (Kiro.dev) ¹⁵ ¹⁶	Claude Sonnet 4; "Auto" mix of models	No	Spec-driven IDE; agent hooks (auto tests/docs); multi-modal chat; multi-file diffs ¹⁵ ¹⁶	✓	Yes	Desktop IDE (Windows/ mac)

Each CLI tool supports running in a terminal (often any OS) and many can use your own model or local LLMs via Model Context Protocol (MCP). For example, Gemini CLI is fully open-source and lets you connect to local or cloud LLMs ¹. Claude Code (Anthropic's agent) maps entire codebases and integrates with GitHub/ GitLab in your CLI ³. GitHub Copilot CLI is closed-source and comes with Copilot subscription, executing agentic tasks using repo context ⁵ ⁶. OpenAI's Codex CLI is open-source, runs GPT-5-Codex locally, and even supports image inputs and web search ⁷ ⁸. Tools like Aider and Kilo are open and BYO-friendly (you supply API keys); Aider focuses on in-terminal pair programming ⁹, while Kilo offers rich agent orchestration (Debugger mode, fail-recovery, parallel agents) ¹³ ¹². Augment Code provides both local IDE agents and cloud-based autonomous agents with advanced context retrieval ¹⁰. Kiro is a closed, spec-driven AI IDE with built-in agentic workflows (e.g. auto-generating tests on save) ¹⁵.

VS Code / IDE Extensions

- **AI Extensions in VS Code:** Embedding AI into popular editors (e.g. VS Code, JetBrains) is widespread. For example, **Roo Code** is an open-source VS Code extension (free, uses your API keys) that turns your editor into a powerful AI “dev team.” It offers lightning-fast multi-file edits, semantic code search, task orchestration modes, and browser automation ¹⁷ ¹⁸ .



Roo Code (open-source VS Code extension) provides specialized AI modes (Code/Architect/Debug), multi-file edits, and permissioned terminal commands ¹⁷ ¹⁸ .

Tool/Extension	Source	Models	BYO Models	Features	Targets	Pricing
GitHub Copilot	Closed	OpenAI GPT (Copilot engine)	No (GitHub service)	Autocomplete, chat, code-completion, multi-line suggestions; deep GitHub integration	VS Code, JetBrains (via plugin)	Subscription (starts \$10/mo)
TabNine	Closed	Codex/LLMs (private)	No	AI code completion & ranking; local inference (paid)	VS Code, JetBrains, etc.	Freemium (Pro/\$20/mo)
CodeWhisperer	Closed (AWS)	AWS LLM (proprietary)	No	Contextual code suggestions, security scanning; supports AWS APIs ¹⁹	VS Code, JetBrains, IntelliJ, AWS Cloud9	Free tier; paid for teams

Tool/Extension	Source	Models	BYO Models	Features	Targets	Pricing
Codeium (Windsurf)	Closed/ Free	Codeium AI (frontier models)	No (cloud or self-host)	Autocomplete, code generation, chat; supports 70+ languages ²⁰	VS Code, JetBrains, Sublime, etc.	Free for individuals; paid enterprise
Roo Code	Open ¹⁷	OpenAI GPT-4o, Claude Sonnet, Gemini, local LLMs ¹⁸	Yes (user API keys)	Multi-agent modes (Code/ Architect/ Debug); fast multi-file diffs; context indexing; browser automation ^{18 21}	VS Code only (local plugin)	Free (open-source)
Kilo Code	Open ¹²	Any (500+ via OpenRouter)	Yes (OpenRouter/ BYO)	Agent modes (Architect, Debug, Orchestrator); context search; automatic recovery from errors ^{13 12}	VS Code, JetBrains (plugin)	Free (open-source)
Cursor	Closed	GPT-4, Claude 3.5 Sonnet, etc. ²²	No (service)	Fork of VS Code with AI modes; multi-file “Composer” edits, in-line assistant, chat; Git integration ^{23 24}	Standalone Editor (VSCode-like UI)	Subscription (e.g. \$20/mo)

Tool/Extension	Source	Models	BYO Models	Features	Targets	Pricing
Continue.dev	Open ²⁵	Any (configurable; local or cloud)	Yes (local/ remote)	Framework for custom assistants: code chat/ completion, prompt “blocks”, agent hub; VS Code/ JetBrains plugins ²⁵	VS Code, JetBrains	Free (OSS)

VS Code and other IDEs support many AI assistants. GitHub Copilot (closed) adds AI completion and chat inside VS Code or JetBrains, but requires a Copilot subscription. AWS’s CodeWhisperer (closed) provides multilingual code suggestions and on-the-fly security checks ¹⁹. TabNine (closed) offers AI completion (including local inference in paid Pro) but is mostly single-line completion. Codeium’s Windsurf plugins (closed) integrate autocomplete and chat with a large language model, free for personal use ²⁰. In contrast, **Roo Code** is fully open-source and local: it adds “agentic” modes to VS Code, allows connecting any model via API, and even runs safe terminal commands ¹⁷ ¹⁸. **Kilo Code** similarly is open and runs as a VS Code/JetBrains extension, offering multiple AI agent modes, memory of past context, and a transparent pricing model (no hidden fees) ¹³ ¹². **Cursor** (a fork of VS Code) is closed-source; it features multi-file editing (its Composer mode), code-chat, and integrated bug-fixing, but locks you into its service. **Continue.dev** is an OSS framework rather than a single assistant – it provides VS Code/JetBrains plugins plus a “hub” to share custom AI helpers; it supports any model (local or API) and is highly modular ²⁵.

AI-Native Editors / IDEs

Editor/ Environment	Source	Built-in Models	BYO Models	AI Features	Platforms
Zed	Open ²⁶	Zeta (open), plus any agent	Yes (select agent)	GPU-accelerated Rust editor; agentic editing (collab human↔AI); in-line AI transforms; multi-model support ²⁶ ²⁷	macOS/ Linux/Win
Cursor Editor	Closed	GPT-4, Claude, etc.	No	VSCode-based AI IDE: real- time completions, multi-file “Composer” mode, chat threads, semantic code search ²³ ²⁸	macOS/ Win/Linux

Editor/ Environment	Source	Built-in Models	BYO Models	AI Features	Platforms
Windsurf Editor	Closed	Cascade agent (frontier mix)	No	Agentic IDE from Codeium: deep codebase embeddings ("Cascade"); live previews & deploy; smart autocomplete (Tab); inline/terminal commands ²⁹ ³⁰	Windows/ macOS/ Linux
Kiro	Closed	Claude Sonnet 4; "Auto" (LLMs)	No	Spec-driven AI IDE: generates specs/tasks from prompts; agent hooks (auto- tests/docs on save); visual diff review; multi-modal input (image prompts) ¹⁵ ¹⁶	Desktop (macOS/ Win)
Augment Code	Closed	Frontier LLMs (cloud service)	No	AI IDE + cloud agents: codebase-aware autocomplete; run AI agents locally or remotely; CLI/TUI interface; context-indexing; image inputs; customizable memories/rules ³¹ ³²	VS Code, JetBrains, Vim, Neovim

Zed and Windsurf (Codeium) exemplify **AI-native IDEs**. Zed is fully open-source (Rust/GPU) and designed for AI collaboration: it supports agentic editing, an in-line assistant, and its own LLM ("Zeta") for on-the-fly predictions ²⁶ ²⁷. Codeium's Windsurf Editor (closed) brands itself as the first "agentic IDE": its Cascade engine embeds entire codebases for deep context, offers one-key tab autocomplete and live site previews, and even auto-fixes linter errors ²⁹ ³³. Cursor Editor (closed) is another AI IDE (built on VS Code) with advanced autocomplete, multi-file "Composer" mode, and an integrated bug-fixer agent ²³ ²⁸. Kiro (closed) is a newer AI IDE focusing on structure: it turns natural-language prompts into specs and tasks, auto-generates tests/docs, and shows code diffs in real time ¹⁵ ¹⁶. Augment (closed) offers AI agents either embedded in the IDE or in the cloud: it provides codebase-aware completions, image/term context, agentic workflows, and the ability to run terminal commands on approval ³¹ ³².

Notes

- **Open vs Closed:** Many enterprise tools (Copilot, Claude Code, Cursor, Kiro, etc.) are closed-source SaaS with fixed model backends. Open-source projects (Zed, Codex CLI, Aider, Kilo, Continue, Goose) allow self-hosting and BYO models, giving more control and privacy ³⁴ ³⁵.
- **BYO Model Support:** Open tools often support any model via APIs. For example, Zed can use various agents (Gemini, Claude, local LLMs), and Roo Code/Kilo rely on your API keys (OpenRouter) ²⁶ ¹². Closed tools typically lock you to their model stack (e.g. Copilot CLI uses GitHub's models only).
- **Multi-file & Project Context:** Open frameworks like Aider and Continue are noted for handling entire codebases and multi-file edits (Aider 'works directly' on your code with full context ⁹, and Thoughtworks praises them for editing across files ³⁶). Closed tools vary in this ability.

- **Models and Context Windows:** Tools differ in context capacity. Google's Gemini CLI offers up to a 1M-token window (exceptional for large repos) ³⁷ . Many tools (Claude, Cursor, etc.) can handle very large code files via sophisticated chunking. Model choice (and any built-in special models) is noted for each tool above.
- **Collaboration & Extra UI:** Some editors add collaboration (Zed has shared screens/chats) ³⁸ , or visual tools (Windsurf's live preview and deploy ³⁹). The tables focus on core coding features, but many environments also include GUI for chat, settings, version control, etc.

Overall, the landscape is rich. Closed-source "agentic" IDEs focus on turnkey productivity and deep integration (e.g. Windsurf, Kiro), while open-source alternatives emphasize flexibility, transparency and BYO models (e.g. Zed, Aider, Kilo, Continue). Each tool's detailed features are cited above for reference ³ ²⁶ ²⁵ .

¹ Gemini CLI | Gemini Code Assist | Google for Developers

<https://developers.google.com/gemini-code-assist/docs/gemini-cli>

² ²⁸ ³⁷ Claude Code vs. Gemini CLI vs. Cursor vs. Qwen Code — Comparing Top AI Coding Assistant | by Fendy Feng | Medium

<https://medium.com/@fendylike/top-ai-coding-assistants-claude-code-vs-gemini-cli-vs-cursor-vs-qwen-code-0bc759fc9d45>

³ ⁴ Claude Code | Claude

<https://claude.com/product/claude-code>

⁵ ⁶ GitHub Copilot CLI · GitHub

<https://github.com/features/copilot/cli>

⁷ ⁸ Codex CLI

<https://developers.openai.com/codex/cli/>

⁹ Aider - AI Pair Programming in Your Terminal

<https://aider.chat/>

¹⁰ ¹¹ ³¹ ³² Augment Code - AI coding platform for real software.

<https://www.augmentcode.com/>

¹² ¹³ ¹⁴ Kilo Code - The best AI coding agent for VS Code and JetBrains

<https://kilocode.ai/>

¹⁵ ¹⁶ Kiro: The AI IDE for prototype to production

<https://kiro.dev/>

¹⁷ ¹⁸ ²¹ Roo Code - Replit Alternatives

<https://replit-alternatives.org/ide-extensions/roo-code/>

¹⁹ Amazon CodeWhisperer: AI-Powered Code Generation - AWS

<https://aws.amazon.com/awstv/watch/50a3d784916/>

²⁰ ²⁵ ³⁴ ³⁵ ³⁶ Best AI Coding Assistants as of November 2025 | Shakudo

<https://www.shakudo.io/blog/best-ai-coding-assistants>

²² ²³ ²⁴ 10 Best AI Coding Tools 2025: Vibe Coding Tools Compared (GitHub Copilot vs Cursor) - Superframeworks Blog

<https://superframeworks.com/blog/best-ai-coding-tools>

26 27 38 Zed — The editor for what's next
<https://zed.dev/>

29 30 33 39 Windsurf Editor | Windsurf
<https://windsurf.com/editor>